



Mechanics

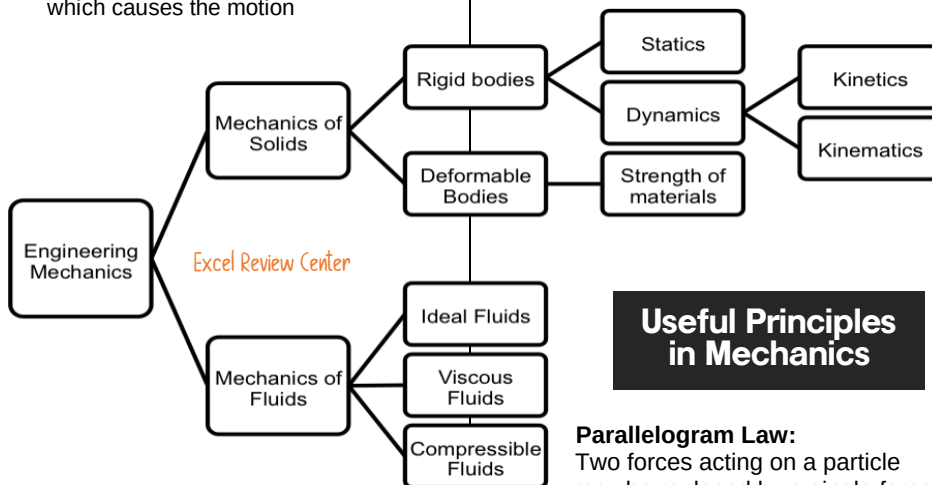
The branch of physics that considers the action of forces on bodies or fluids that are both at rest and in motion.

Engineering Mechanics

The branch of engineering that applies the principles of mechanics to any design that must take into account the effect of forces.

Branches of Mechanics of Solids:

- Statics** : It deals with bodies in the state of rest.
- Kinetics** : It deals with bodies in motion under the action of forces.
- Kinematics** : It refers to the study of motion without reference to the forces which causes the motion



Useful Principles in Mechanics

Parallelogram Law:
Two forces acting on a particle may be replaced by a single force (resultant) which can be obtained by drawing the diagonal of a parallelogram, which has the sides equal to the given forces.

Principle of Transmissibility:

The conditions of equilibrium or motion of a rigid body remains unchanged if a force acting at a given point of the rigid body is replaced by a force of the same magnitude and direction, but acting at a different point provided that the two forces have the same line of action.

Resolution of Forces into Components:

It is the process of replacing a single force acting on a particle by two or more forces that have the same effect as that of a single force.

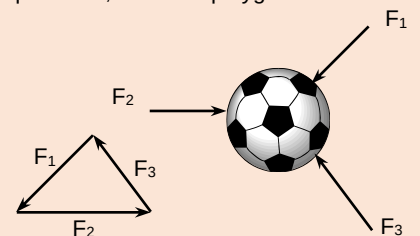
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Conditions for Equilibrium

A body is in equilibrium when both the resultant force and the sum of the moments of all external forces about any random point are equal to zero. if it satisfies the following conditions.

Graphical Method

Under this condition, the forces or vectors are transformed into a force polygon. For equilibrium, the force polygon must close.

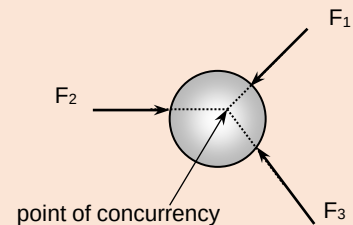


Force Polygon

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Directional Method

If three or more non-parallel forces or vectors are in equilibrium, then they must be concurrent.



Analytical Method

If forces or vectors are in equilibrium, then it must satisfy the three static equations; namely

$$\sum F_x = 0 \quad \longrightarrow = \longleftarrow$$

$$\sum F_y = 0 \quad \uparrow = \downarrow$$

$$\sum M_x = 0 \quad \curvearrowright = \curvearrowleft$$

History of Mechanics

Scientist	Area of Development
Aristotle	Lever, Center of gravity, basic mechanics
Archimedes	Buoyancy
Galileo	Mathematical analysis, experimentation
Kepler	Planetary motion, Kepler's laws
Newton	Laws of Motion, Law of Gravitation
Bernoulli	Principle of Virtual Work
Euler	Rigid bodies, moment of inertia, momentum
D'alembert	Dynamic equilibrium
Planck and Bohr	Quantum Mechanics